MATH3018 Assignment 6, due on Apr 7

(1) (4pts) Let $X = (X_1, X_2, X_3)$ have a population covariance matrix

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma^2 & \sigma^2\rho & 0 \\ \sigma^2\rho & \sigma^2 & \sigma^2\rho \\ 0 & \sigma^2\rho & \sigma^2 \end{pmatrix}, -\frac{1}{\sqrt{2}} < \rho < \frac{1}{\sqrt{2}}$$

Find the principal components and the variance explained by each.

[**Solution**]

To find the eigenvalues, we let $\det(\mathbf{\Sigma} - \lambda \boldsymbol{I}) = 0$, i.e.,

$$\begin{vmatrix} \sigma^2 - \lambda & \sigma^2\rho & 0 \\ \sigma^2\rho & \sigma^2 - \lambda & \sigma^2\rho \\ 0 & \sigma^2\rho & \sigma^2 - \lambda \end{vmatrix} = 0$$

By solving the system, we obtain the characteristic polynomial in terms of λ as:

$$\left(\sigma^2 - \lambda\right)\left(\sigma^4 - 2\sigma^4\rho^2 - 2\lambda\sigma^2 + \lambda^2\right) = 0$$

and we get $\lambda_1 = \sigma^2(1 + \sqrt{2}\rho), \lambda_2 = \sigma^2$, and $\lambda_3 = \sigma^2(1 - \sqrt{2}\rho)$. To solve the eigenvector, we need to solve $\Sigma \boldsymbol{e}_i = \lambda_1 \boldsymbol{e}_i$, for $i = 1, 2, 3$. We found that

$$\boldsymbol{e}_1 = \begin{pmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{pmatrix}, \boldsymbol{e}_2 = \begin{pmatrix} \sqrt{2}/2 \\ 0 \\ -\sqrt{2}/2 \end{pmatrix}, \quad \text{and } \boldsymbol{e}_3 = \begin{pmatrix} 1/2 \\ -\sqrt{2}/2 \\ 1/2 \end{pmatrix}$$

Therefore, the principal components become

$$\begin{aligned} Y_1 &= \boldsymbol{e}_1^T \boldsymbol{X} = \frac{1}{2}X_1 + \frac{\sqrt{2}}{2}X_2 + \frac{1}{2}X_3 \\ Y_2 &= \boldsymbol{e}_2^T \boldsymbol{X} = \frac{\sqrt{2}}{2}X_1 - \frac{\sqrt{2}}{2}X_3 \\ Y_3 &= \boldsymbol{e}_3^T \boldsymbol{X} = \frac{1}{2}X_1 - \frac{\sqrt{2}}{2}X_2 + \frac{1}{2}X_3 \end{aligned}$$

The total population variance is

$$\sum_{i=1}^{3} \operatorname{Var}(Y_i) = \sum_{i=1}^{3} \lambda_i = \sigma^2(1 + \sqrt{2}\rho) + \sigma^2 + \sigma^2(1 - \sqrt{2}\rho) = 3\sigma^2$$

and the variance explained by each principal components is: $\frac{1}{3}(1 + \sqrt{2}\rho)$, $\frac{1}{3}$, and $\frac{1}{3}(1 - \sqrt{2}\rho)$, for Y_1, Y_2, and Y_3, respectively.

(2) (7pts) $n = 10$ observations from a bivariate population are listed as follows.

(a) Using sample covariance matrix, determine the sample principal components Y_1 and Y_2, and their variances.

(b) Find the proportion of the total sample variance explained by Y_1.

(c) Compute the correlation coefficients $\operatorname{Cor}(Y_1, X_j)$, for $j = 1$ and 2 respectively.

(d) Let $Z = (Z_1, Z_2)^T$ denote the standardized variables. Determine the sample principal components Y_1^Z and Y_2^Z, and their variances.

Observation #	X_1	X_2
1	108.28	17.05
2	152.36	16.59
3	95.04	10.91
4	65.45	14.14
5	62.97	9.52
6	263.99	25.33
7	265.19	18.54
8	285.06	15.73
9	92.01	8.10
10	165.68	11.13

(e) Find the proportion of the total sample variance explained by Y_1^Z.
(f) Compute the correlation coefficients $\mathrm{Cor}\left(Y_1^Z, Z_j\right)$ for $j = 1$ and 2 respectively. Comment on Y_1^Z.
(g) Comment on your findings comparing Questions (c) and (f).

[Solution]

(a) Sample covariance matrix is

$$S = \begin{pmatrix} 7476.45 & 303.62 \\ 303.62 & 26.19 \end{pmatrix}$$

S has eigenvalues $\lambda_1 = 7488.80$ and $\lambda_2 = 13.84$, with corresponding eigenvectors $u_1 = (0.999, 0.0407)^T$ and $u_2 = (0.0407, -0.999)^T$. Therefore $Y_1 = u_1^T X = 0.999X_1 + 0.0407X_2$ and $Y_2 = u_1^T X = 0.0407X_1 - 0.999X_2$. $\mathrm{Var}\left(Y_1\right) = \lambda_1 = 7488.80$ and $\mathrm{Var}\left(Y_2\right) = \lambda_2 = 13.84$.

(b) The proportion of variance explained by Y_1 is $\lambda_1 / \left(\lambda_1 + \lambda_2\right) = 0.998$.

(c) $\mathrm{Cor}\left(Y_1, X_1\right) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = 0.9998$ and $\mathrm{Cor}\left(Y_1, X_2\right) = u_{12}\sqrt{\lambda_1}/\sqrt{s_{22}} = 0.6882$. Y_1 is almost completely determined by X_1 due to its large sample variance relative to X_2.

(d)

$$R = \begin{pmatrix} 1 & 0.6861 \\ 0.6861 & 1 \end{pmatrix}$$

R has eigenvalues $\lambda_1^Z = 1.6861$ and $\lambda_2^Z = 0.3139$, with corresponding eigenvectors $u_1 = (0.7071, 0.7071)^T$ and $u_2 = (0.7071, -0.7071)^T$. Therefore $Y_1^Z = u_1^T Z = 0.7071Z_1 + 0.7071Z_2$ and $Y_2^Z = u_2^T Z = 0.7071Z_1 - 0.7071Z_2$. $\mathrm{Var}\left(Y_1^Z\right) = \lambda_1^Z = 1.6861$ and $\mathrm{Var}\left(Y_2^Z\right) = \lambda_2^Z = 0.3139$.

(e) The proportion of variance explained by Y_1^Z is $\lambda_1^Z / \left(\lambda_1^Z + \lambda_2^Z\right) = 0.8431$.

(f) $\mathrm{Cor}\left(Y_1^Z, Z_1\right) = u_{11}\sqrt{\lambda_1^Z} = 0.918$ and $\mathrm{Cor}\left(Y_1^Z, Z_2\right) = u_{12}\sqrt{\lambda_1^Z} = 0.918$. Z_1 and Z_2 contribute to Y_1^Z equally.

(g) Quite different results are obtained in Questions (c) and (f), mostly because $\mathrm{Var}\left(X_1\right) >> \mathrm{Var}\left(X_2\right)$. In the application of PCA, we often want to standardize the data, i.e. to use the sample correlation matrix rather than sample covariance matrix, to remove the influence of differing variable scales.

(3) (4pts) The weekly rates of return for five stocks listed on the New York Stock Exchange are given in Table 1 below. JPMorgan, CitiBank and WellsFargo are financial service companies; RoyDutShell and ExxonMobile are energy companies. (the full dataset "stocks-dat" has been uploaded to Moodle).

TABLE 1. Stock-price data (weekly rate of return)

Week	JPMorgan	CitiBank	WellsFargo	RoyDutShell	ExxonMobil
1	0.0130338	-0.0078431	-0.0031889	-0.0447693	0.0052151
2	0.0084862	0.0166886	-0.0062100	0.0119560	0.0134890
3	-0.0179153	-0.0086393	0.0100360	0.0000000	-0.0061428
⋮	⋮	⋮	⋮	⋮	⋮
102	0.0103929	-0.0026612	0.0044290	-0.0024818	-0.0164502
103	-0.0127948	-0.0143678	-0.0187402	-0.0049759	-0.0163732

(a) Construct the sample covariance matrix S, and find the sample principal components.
(b) Determine the proportion of the total sample variance explained by the first three principal components. Interpret these components.
(c) Construct Bonferroni simultaneous 90% confidence intervals for the variances λ_1, λ_2, and λ_3 of the first three population components Y_1, Y_2, and Y_3.
(d) Given the results in Parts (a)-(c), do you feel that the stock rates-of-return data can be summarized in fewer than five dimensions? Explain.

[Solution]

(a) The sample covariance matrix S is shown below:

	JPMorgan	CitiBank	WellsFargo	RoyDutShell	ExxonMobil
JPMorgan	0.00043327	0.00027567	0.00015903	0.00006412	0.00008897
CitiBank	0.00027567	0.00043872	0.00017997	0.00018145	0.00012326
WellsFargo	0.00015903	0.00017997	0.00022397	0.00007341	0.00006055
RoyDutShell	0.00006412	0.00018145	0.00007341	0.00072250	0.00050828
ExxonMobil	0.00008897	0.00012326	0.00006055	0.00050828	0.00076567

The PCs loadings are given as:

	PC1	PC2	PC3	PC4	PC5
JPMorgan	-0.2228228	0.6252260	-0.32611218	0.6627590	-0.11765952
CitiBank	-0.3072900	0.5703900	0.24959014	-0.4140935	0.58860803
WellsFargo	-0.1548103	0.3445049	0.03763929	-0.4970499	-0.78030428
RoyDutShell	-0.6389680	-0.2479475	0.64249741	0.3088689	-0.14845546
ExxonMobil	-0.6509044	-0.3218478	-0.64586064	-0.2163758	0.09371777

(b) From part (a),

$$\hat{\lambda}_1 = 0.00137, \hat{\lambda}_2 = 0.00070, \hat{\lambda}_3 = 0.00025, \hat{\lambda}_4 = 0.00014, \hat{\lambda}_5 = 0.00012,$$

so the total sample variance is $\sum_{i=1}^{5} \hat{\lambda}_i = 0.00258$ and the proportion of total variance explained by the first three component is $\sum_{i=1}^{3} \hat{\lambda}_i / \sum_{i=1}^{5} \hat{\lambda}_i = 0.8988$. The first component might be interpreted as a "market" component with the greatest weight (in absolute value) on Royal Dutch Shell and Exxon Mobil, the second component as an "industry" component that separates bank and gas companies.

and the third component is a contrast between these five stocks which is difficult to interpret.

(c) The Bonferroni simultaneous $100(1-\alpha)\%$ confidence interval for λ_i can be constructed by

$$\frac{\hat{\lambda}_i}{1+z\left(\frac{\alpha}{2m}\right)\sqrt{\frac{2}{n}}} \le \lambda_i \le \frac{\hat{\lambda}_i}{1-z\left(\frac{\alpha}{2m}\right)\sqrt{\frac{2}{n}}}.$$

Thus the 90% confidence intervals for the three variance of the population components are:

$$\lambda_1 : (0.001055, 0.001944)$$
$$\lambda_2 : (0.000541, 0.000997)$$
$$\lambda_3 : (0.000196, 0.000361)$$

(d) Stock returns are probably best summarized in two dimensions with 80% of the total variation accounted for by a "market" component and an "industry" component without much loss of information.